Ex.no.2
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Titanic Dataset Analysis and Preprocessing using Pandas and SimpleImputer

AIM:

To load Titanic dataset, handle missing values using simple imputer, & prepare the data for training & testing.

PROCEDURE:

1. Load Titanic Dataset
2. Explore dataset (shape, info, stats).
3. Handle missing values (Simple Imputer)
4. Visualize passenger class distribution
5. Filter/analyze data
6. Split dataset into training & testing sets

CODE:

```
import pandas as pd, seaborn as sns, matplotlib.pyplot as plt
from sklearn.impute import SimpleImputer
from sklearn.model_selection import train_test_split
df = sns.load_dataset('titanic')
df['age'] = SimpleImputer(strategy='mean').fit_transform(df[['age']])
df['deck'] = df['deck'].fillna('Unknown')
df['embarked'] = df['embarked'].fillna(df['embarked'].mode()[0])
sns.countplot(x='pclass', data=df)
plt.title("Passenger Class Distribution")
plt.show()
```

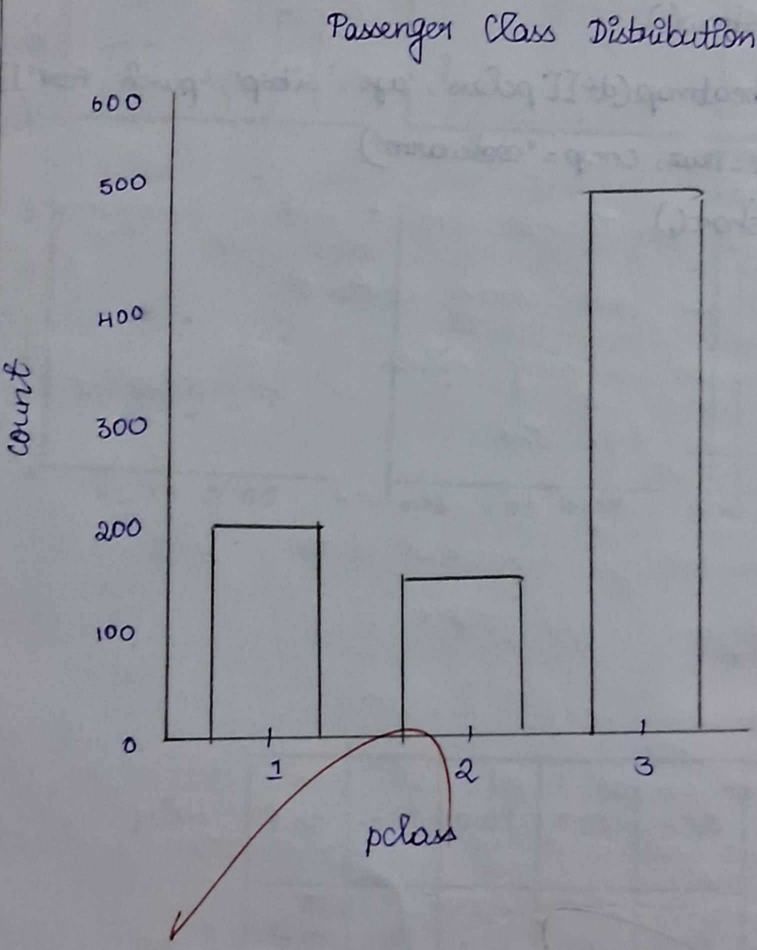
```
print("Females who survived:", df[(df.sex=='female') &
(df.survived == 1)].index.tolist())
print("3rd class under 16:", df[(df.pclass==3)& (df.age<16)].index.tolist())
print("Embarked 'C' & class 2:", df[df.emarked == 'C')&
(df.pclass==2)].index.tolist())
print("Top 5 oldest survivors:", df[df.survived==1].nlargest(5,
'age')['age'].tolist())
print("Zero fare passengers:", df[df.fare==0].index.tolist())

train, test= train_test_split(df, test_size=0.2, random_state=42)
print("Train size:", len(train), "\nTest size:", len(test))
```

RESULT:

Thus the program successfully analysed the data queries, splitted the dataset, and prepared it for modelling.

OUTPUT:



Passenger Class Distribution
600
500
400
300
200
100
0
count
1
2
3
pclass

Females who survived : 177 passengers

3rd class under 16 : 73 passengers

Embarked 'C' & class 2 : 66 passenger

Top 5 oldest survivors : 80, 63, 63, 62, 62 years

Zero fare passengers : 70 passengers

Train size : 712

Test size : 179